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Research Article

Revealing the effect of alumina addition on the residual stress in glass-to-metal seals *via* photoluminescence spectroscopy

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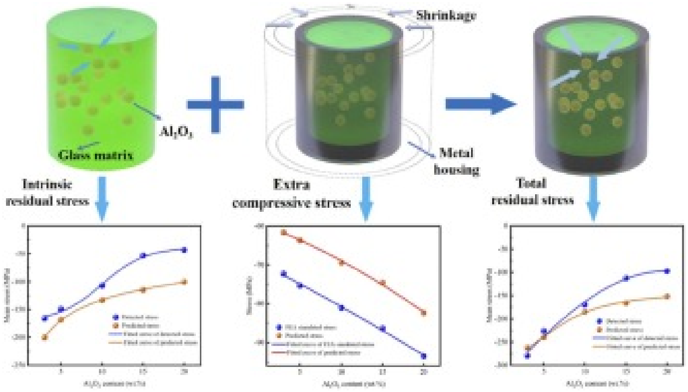
Highlights

- A novel approach using photoluminescence spectroscopy to detect residual stress in glass-to-metal seals.
- Increasing alumina in glass composites lowers residual stress.
- Typesetting math: 100% ates the applicability of strain energy theory to explain stress behavior in glass-to-metal seals.

Abstract

In Glass-to-Metal (GTM) seals, glass matrix composites (GMCs) are frequently utilized to enhance sealing performance. However, the incorporation of reinforcements inevitably impacts the residual stress, a critical determinant of the hermeticity of GTM seals. Hence, revealing the effect of reinforcements on the residual stress is urgently needed. This study investigates the correlation between residual stress and varying alumina content. Theoretically, the total residual stress in GTM seals comprises both intrinsic residual stress and extra compressive stress. The former is due to the thermal mismatch between the glass matrix and reinforcements, while the latter arises from the cooling-induced shrinkage of the metal housing. To empirically validate this theoretical framework, Cr^{3+}-doped Al_2O_3 particles were incorporated into borosilicate glass to prepare GMCs with alumina contents ranging from 3 wt% to 20 wt%. Leveraging the stress-sensitive photoluminescence property, photoluminescence spectroscopy (PLS) was applied to experimentally measure stress, enabling a detailed examination of stress distribution within GTM seals. Concurrently, finite element modeling was employed to estimate the extra compressive stress imposed by the metal housing. The experimentally detected stresses closely align with the theoretical predictions, thus verifying the accuracy of the theoretical framework. A significant finding is the observed reduction in residual stress within GTM seals with increasing alumina content. This trend is explained through strain energy theory, offering comprehensive insights into the influence of alumina content on the residual stress. The findings provide a robust foundation for the design and optimization of the stress in GTM seals.

Graphical Abstract



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Keywords

Glass-to-metal seal; Glass matrix composites; Residual stress; Strain energy; Photoluminescence spectroscopy

1. Introduction

Glass-to-Metal (GTM) seals are renowned for their exceptional hermeticity and pronounced electrical insulation properties, making them indispensable in a wide range of electrical and electronic devices [1]. These seals are extensively employed in electrical penetration assemblies (EPAs) [2], solid oxide fuel cells [3], aerospace applications [4], and medical implants [5], with EPAs, in particular, receiving significant recent attention [6]. The operational demands on EPAs require them to maintain high reliability and impeccable hermeticity under extreme conditions, including elevated temperatures, substantial pressures, and

t gas or liquid transfer between environments [7], [8]. This mandates the sealing components to possess both robust bonding strength and hermeticity. However, ensuring sustained hermeticity, especially during prolonged exposure to high temperatures, remains a significant challenge [9]. Preliminary studies have indicated a strong correlation between the sealing performance of GTM seals and the distribution of residual stress within them [10]. Excessive residual stress can induce the formation of cracks and other structural defects, while insufficient residual stress may result in hermeticity failure [5], [7], [11], [12]. Hence, accurately determining residual stresses is essential for optimizing seal performance.

Moreover, to enhance the stability of GTM seals, the focus of research has shifted towards the application of glass matrix composites (GMCs) that incorporate key reinforcements such as alumina (Al_2O_3) [10], zirconia (ZrO_2) [13], and silicon carbide (SiC) [14]. Among these, the addition of Al_2O_3 particles into the glass matrix has emerged as a widely adopted approach [15], [16]. While it is generally agreed upon that introducing a second phase into the glass matrix can generate intrinsic residual thermal stresses [17], [18]. As illustrated in Fig. 1, within a GTM seal configuration comprising a GMC material and a metal housing, intrinsic residual stress inevitably arises in the glass matrix due to the mismatch in the coefficients of thermal expansion (CTE) between the glass and the reinforcement particles [19]. Additionally, during the cooling process, the inward shrinkage of the metal housing imposes extra compressive stress on the glass composite [20]. The total residual stress in a GTM seal is thus the resultant of these intrinsic residual stress and extra compressive stress [21]. It is noteworthy that for GTM seals without reinforcements, the total residual stress is predominantly due to the inward compression exerted by the metal housing, namely, the intrinsic residual stress is negligible [22]. What's more, previous studies have primarily focused on the mechanical properties of GMCs within GTM seals [23], with limited systematic investigation into how reinforcements influence the residual stress within the seal. Therefore, revealing the effect of second phase additions on residual stress in GTM seals is essential for advancing knowledge in this area.

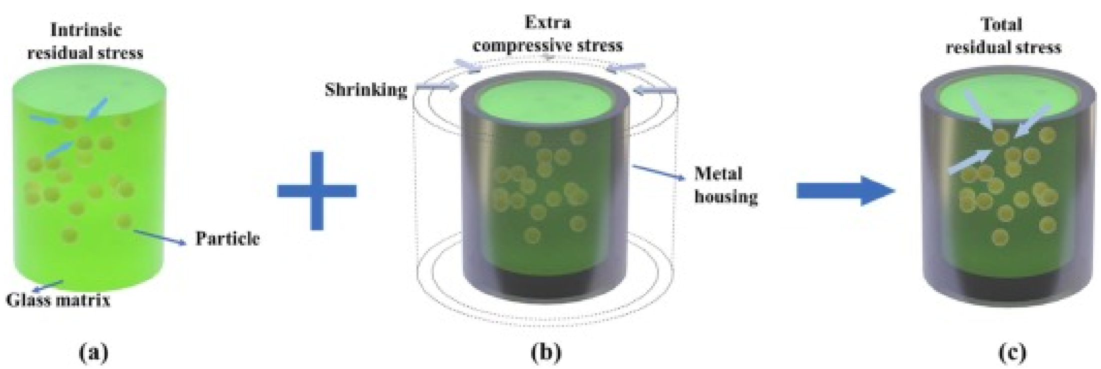

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Fig. 1. Schematic diagram of intrinsic residual stress (a), extra compressive stress (b), and total residual stress (c) in GTM seals.

In the field of GTM seals, variety of methods have historically been employed to detect residual stress, including X-ray diffraction [24], fiber Bragg grating sensors [25], nuclear magnetic resonance [17], indentation techniques [26], and particularly, photoluminescence spectroscopy (PLS) [27]. PLS, which determines stress by detecting frequency shifts in characteristic luminescence lines (either Raman or fluorescence), has proven to be highly efficient in determining residual stress [28], [29], [30], [31], [32]. The underlying principle of the PLS technique relies on the unique optical absorption and emission of a crystallized material in response to strain. Such responses often stem from the localized atomic environment of a host compound surrounding an impurity, as typically described by crystal field theory [33], [34]. The use of Cr^{3+}-doped Al_2O_3 particles has significantly advanced the determination of residual stress in glass composite materials, including in systems such as Al_2O_3 fibers/soda-lime glass [35] and Al_2O_3 platelets/borosilicate glass [36]. It is important to note that the reinforcing particles within the glass matrix are randomly oriented and dispersed throughout the glass substrate, which averages out any crystal orientation dependencies. Therefore, the PLS method primarily measures the deformation caused by hydrostatic stress acting on the particles, and the converted stress values represent the hydrostatic stress [37], [38].

This study primarily aims to investigate the impact of Al_2O_3 particle incorporation on residual stress in GTM seals. Three key innovations were introduced: the novel application of PLS for stress detection, the development of integrated structure/function GMCs system by embedding Cr^{3+}-doped Al_2O_3 particles into a borosilicate glass matrix, and the use of a strain energy perspective to analyze stress. The total residual stress in GTM seals is theoretically regarded as the sum of intrinsic residual stress and extra compressive stress, with corresponding calculation formulas provided. To validate the accuracy of these theoretical predictions, GMCs were then either sealed with 304 stainless steel (304SS) or subjected to a simulated sealing process without metal housing. The PLS technique was employed to measure both intrinsic and total residual stresses, while finite element analysis (FEA) modeling was utilized to estimate the extra compressive stress applied by the metal housing. A comparison of the experimental measurements with theoretical predictions was conducted, revealing important insights. By applying strain energy theory, this analysis provides a comprehensive understanding of how Al_2O_3 particle incorporation influences residual stress in GTM seals, offering valuable theoretical insights and practical guidelines for future seal design and applications.

2. Experimental procedures

2.1. Materials

A commercial borosilicate glass, designated TH-3, sourced from the State Key Laboratory of New Ceramics and Fine Processing in Beijing, was utilized as the matrix material. Al_2O_3 particles doped with 0.16 at% Cr^{3+} (Guangxi Wuzhou Xingyuegem Co., Ltd., Wuzhou, China) were incorporated into the glass matrix to fabricate the glass composites. 304SS metal housing, with the measurements of an inner diameter of 5 mm, an outer diameter of 10 mm, and a height of 5 mm were used to seal with glass composites.

2.2. Sample preparation

The Al_2O_3 particles were mixed with glass powders in concentrations of 3 wt%, 5 wt%, 10 wt%, 15 wt%, and 20 wt%, respectively. Following this, the mixtures were subjected to a ball-milling process in an ethanol medium for 3 h using a vertical planetary ball mill (XQM-2A, Tianchan Powder). Agate jars, each with a capacity of 250 mL, were employed as the milling containers. The grinding medium consisted of ZrO_2 balls, specifically 20 balls of 10 mm diameter, 50 balls of 8 mm diameter, and 400 balls of 5 mm diameter. The ratio of ball to material was maintained at 6:1, with the milling conducted at a rotation speed of 240 r/min. Post-milling, the mixtures were dried and subsequently mixed with an organic binder to form cylindrical green bodies. These were then heated to 680℃for a duration of 3 h to achieve binder burnout, resulting in the final glass preforms.

For the preparation of GTM seals, the 304SS metal housing was ultrasonically cleaned in alcohol for 30 min to remove residues. To ensure chemically bonded seals, these housings were pre-oxidized in a tubular furnace at 1050℃ under a nitrogen flow of 1000 mL/min for 45 min, preventing iron oxide formation [39]. Following pre-oxidation, the 304SS metal housings were paired with one batch of glass preforms, assembled onto a graphite gasket, and sealed in a horizontal tube furnace at 980℃ within a nitrogen atmosphere for 30 min. After sealing, the samples were sequentially sanded with SiC sandpapers of various grits (100, 240, 600, 800, 1500, 2000 and 3000), polished using a 0.5 μm diamond suspension, and ultrasonically cleaned, preparing them for further characterization. Seals with Al_2O_3 particle concentrations of 3 wt%, 5 wt%, 10 wt%, 15 wt%, and 20 wt% were designated as A_3*-seal*, A_5*-seal*, A_{10}*-seal*, A_{15}*-seal*, and A_{20}*-seal*, respectively.

To assess the intrinsic residual stresses within the GMCs, another batch of glass preforms was assembled with graphite molds, whose dimensions were the same as 304SS housing. The graphite molds, with a CTE ≤ 3.5 × 10^{-6}/℃ (100 – 600℃), exhibit a lower CTE than borosilicate glass, preventing significant inward shrinkage and thus avoiding the introduction of extra compressive [illegible]ss composites. Additionally, due to the non-wetting nature of molten glass on graphite surfaces, the molds and samples automatically separate upon cooling, meaning that the graphite molds could not form seals with the GMCs [21]. These samples, devoid of 304SS housing, underwent the same temperature profile and post-processing as the sealed samples, allowing for an accurate assessment of the intrinsic residual stresses generated during the sealing process. The resulting samples were labeled as A_3-*unseal*, A_5-*unseal*, A_{10}-*unseal*, A_{15}-*unseal* and A_{20}-*unseal*, according to the Al_2O_3 particle concentrations, respectively.

2.3. Characterization

2.3.1. Physical properties of the materials

The density of the GMCs was obtained using a gas pycnometer (AccuPyc II 1340, Micromeritics, Georgia, USA). To ascertain the Young's moduli of the materials, following the method reported eleswhere [40], a quasi-static uniaxial compressive test was performed using an electronic universal testing machine (INSTRON 5982, Shanghai, China) at a speed of 0.5 mm/min. The CTE and glass transition temperature (T_g) of the borosilicate GMCs were measured using a dilatometer (Model DIL 402CL, NETZSCH, Germany) with a heating rate set at 5 ℃/min. Table 1 displays the physical and mechanical properties of the materials.

Table 1. Mechanical properties of borosilicate GMCs, Al_2O_3 and 304SS.

Samples		**Density**			
Borosilicate glass		**(g/cm^3)**	**Young’s modulus (GPa)**	**CTE (×10^{-6}℃$^{-1}$) (25–550℃)**	**T_g (℃)**
Al_2O_3 content (wt%)	0	2.51	73	6.88	556.2
	3	2.54	76	6.76	558.4
	5	2.58	78	6.71	561.9
	10	2.62	86	6.65	562.3
	15	2.68	93	6.42	565.6
	20	2.74	102	6.28	567.2
Al_2O_3 particles [41]		3.96	375	5.8	/

Samples	Density			
Borosilicate glass	(g/cm^3)	Young's modulus (GPa)	CTE (×10^{-6}℃$^{-1}$) (25–550℃)	T_g (℃)
304SS [25]	/	193	18.4	/

2.3.2. Stress determination

In order to detect the stress within the glass region, a Raman spectrometer (LabRAM HR, Horiba, France) equipped with an 1800 lines/mm diffraction grating, a high-sensitivity Peltier-cooled charge coupled device (CCD), and a motorized xyz microscope stage was utilized. A laser light source argon ion laser (λ = 532 nm) was used for excitation, while an optical microscope was used both to focus the laser on the sample and to collect the scattered light. The scanning range for all measurements was set between 14,200 cm^{-1} and 14,500 cm^{-1}, with a resolution of 0.3 cm^{-1}. The software LabSpec 6 (LabRAM HR, Horiba, France) was used for peak fitting and to determine peak positions.

For this application, the primary focus is on the shift of the Cr^{3+} emission line (R-line) in Cr^{3+}-doped Al_2O_3 particles embedded within the glass matrix. The fluorescence lines observed at approximately 14,402 cm^{-1} (R1) and 14,433 cm^{-1} (R2) [33], are caused by the crystal field split excitation of 3d electrons from the Cr^{3+} that substituted for Al^{3+} atoms in the Al_2O_3. The shift in the R-line is stress-dependent, allowing for accurate stress calculations by recording the magnitude of the R-line shift [21], [27]. It should be noted that the stress values obtained using the PLS technique represent hydrostatic stress, which is scalar in nature. To better represent the residual stress state in GTM seals, the positive and negative signs were used to indicate tensile and compressive stress, respectively.

2.4. Finite element analysis

FEA simulations were employed to evaluate the impact of extra compressive stress on the GTM seals. A finite element model was established by the commercial finite element software ABAQUS/Standard (version 6.14). Both the glass composites and 304SS were modeled as linear elastic materials. The FEA simulation began at an assumed stress-free temperature of 560℃ and matched the cooling portion of the thermal process. Since the glass and 304SS metal housing were bonded together in the whole process

Typesetting math: 100% osity of the glass, the contact conditions for all the materials were set as ties. To ensure consistency between the initial conditions of the experiment and the corresponding simulation, the parameters employed in the FEA were directly derived from experimental data, as listed in Table 1. The simulation results are presented as von Mises stress values.

3. Results and discussion

3.1. Theoretical prediction of stress

Based on the information presented in Fig. 1, the following relationship among intrinsic residual stress (σ_i), extra compressive stress (σ_e), and total residual stress (σ_t) can be deduced [21]:

$$\sigma_t = \sigma_i + \sigma_e \tag{1}$$

It is noteworthy that, fundamentally, if all factors are considered comprehensively, this equation does not represent a simple arithmetic addition but rather a conceptual combination that reflects the distribution and interaction of these stresses within the material [21]. However, current technological and methodological limitations make it challenging to decouple the complex interactions between stress, material properties, components, and environmental factors. Given that the PLS technique measures hydrostatic stress as a scalar, the equation can be simplified to represent a straightforward summation. In subsequent calculations, the stress values are treated as scalars for clarity and consistency.

Intrinsic residual stress can be calculated using the model proposed by Hsueh and Becher et al. [18]:

$$\sigma_i = \frac{(\alpha_A - \alpha_g)\Delta T}{\frac{1}{3K_A} + \frac{1}{4(1-f)G_g} + \frac{f}{3(1-f)K_g}} \tag{2}$$

Where, α_A and α_g represent the CTE of the Al_2O_3 particles and the glass matrix, respectively. The bulk moduli of the Al_2O_3 particles and the glass matrix are symbolized as K_A and K_g, respectively. G_g is the shear moduli of glass matrix. The parameter f corresponds to the volume fraction of Al_2O_3 particles, which can be calculated based on the densities of the respective systems. ΔT is the temperature range during which stress evolution remains unaffected by inelastic processes as the material cools. The temperature at which the glass matrix becomes too viscous to relax its thermal expansion mismatch with the particles is defined as T_g [42].

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The bulk moduli K and shear moduli G can be derived as follows:

$$K = \frac{E}{3\times(1-2\nu)} \tag{3}$$

$$G = \frac{E}{2\times(1+\nu)} \tag{4}$$

Where, E denotes Young's modulus. ν represents the Poisson's ratio, with values of 0.33 for glass, 0.41 for alumina, and 0.29 for 304SS [21].

To calculate the extra compressive stress, an incompatible strain ε_{in} is considered, which is due to a small temperature change in ΔT. The expressions are given as follows [43], [44]:

$$\Delta\alpha = (\alpha_{gc} - \alpha_m) \tag{5}$$

$$\varepsilon_{in} = \Delta\alpha(T_r - T_g) \tag{6}$$

Where, $\Delta\alpha$ is the difference in the true CTE between the glass composites (α_{gc}) and the metal housing (α_m). The room temperature is denoted by T_r. A more precise definition of ε_{in} could be described as follows since the CTE vary with changes in temperature [45]:

$$\varepsilon_{in} = \int_{T_g}^{T_r} (\alpha_{gc} - \alpha_m)\, dT \tag{7}$$

Hence, σ_e can be derived as follows:

$$\sigma_e = E_{gc} \times \varepsilon_{in} = E_{gc} \times \int_{T_g}^{T_r} (\alpha_{gc} - \alpha_m)\, dT \tag{8}$$

3.2. Experimental results

3.2.1. Intrinsic residual stress

The schematic diagram in Fig. 2a illustrates the setup for PLS detection used to measure the Raman shift in the glass composites. Typesetting math: 100% points strategically selected on the flat surface of the sample with a step size of 10 μm were used to collect the Raman fluorescence spectrum at each point. The measured values were arithmetically averaged to determine the mean stress (

σ_m) in the sample. Fig. 2b-f present a comparison of residual stress values in GMCs with varying Al_2O_3 particle ratios. The data reveal that as the Al_2O_3 particle content increases, the stress fluctuations diminish, leading to progressively stabilized measurements that converge to a fixed value. For instance, in the A_3*-unseal* and A_5*-unseal* samples, significant fluctuations in stress are observed, whereas in the A_{10}*-unseal*, A_{15}*-unseal*, and A_{20}*-unseal* samples, the stress fitting results are represented by nearly straight lines, indicating minimal variation and greater stability in the stress measurements.

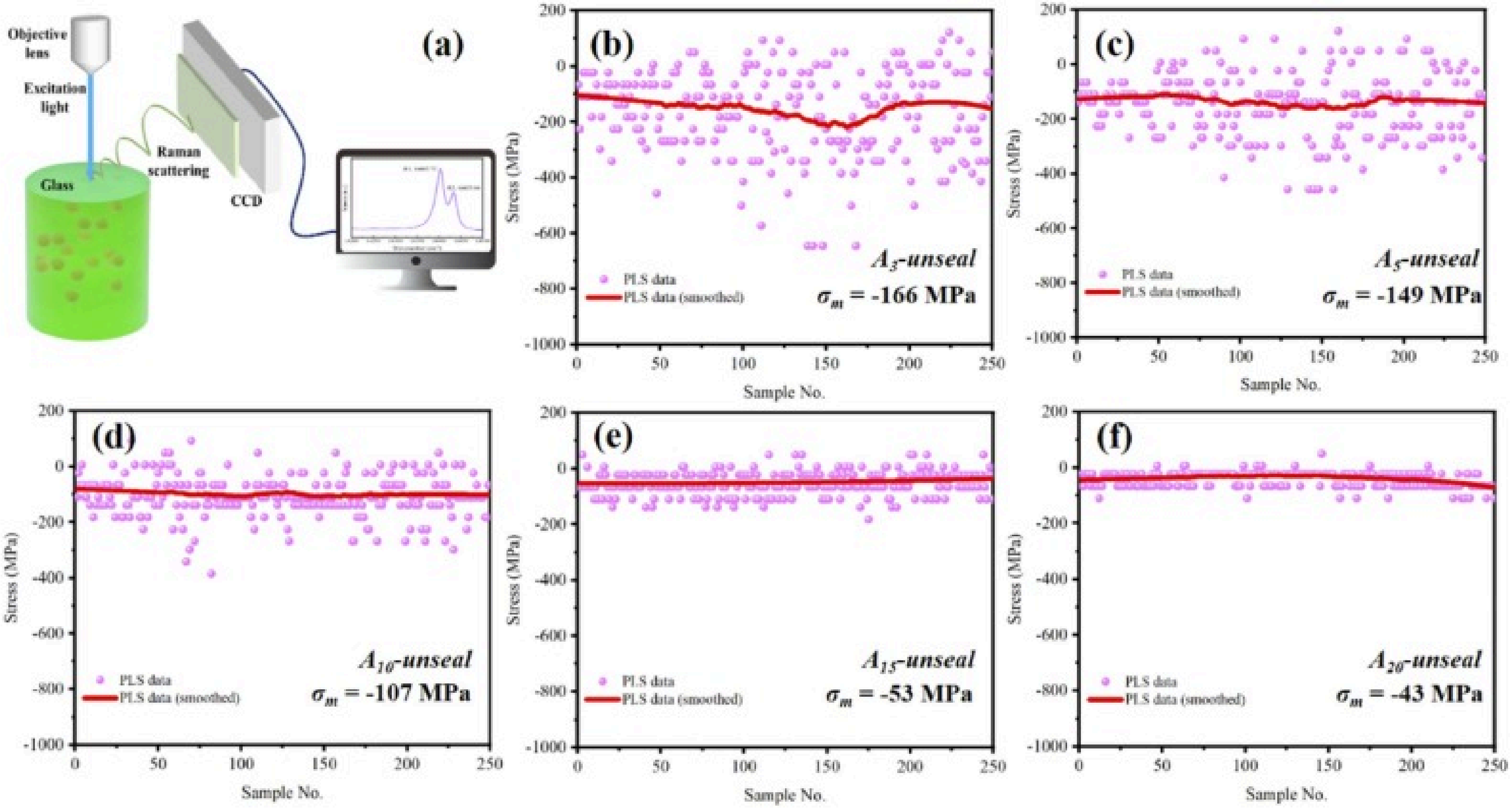

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Fig. 2. (a) Schematic illustration for PLS detection, (b) – (f) present the intrinsic residual stress obtained by the PLS technique for A_3*-unseal*, A_5*-unseal*, A_{10}*-unseal*, A_{15}*-unseal* and A_{20}*-unseal*, respectively.

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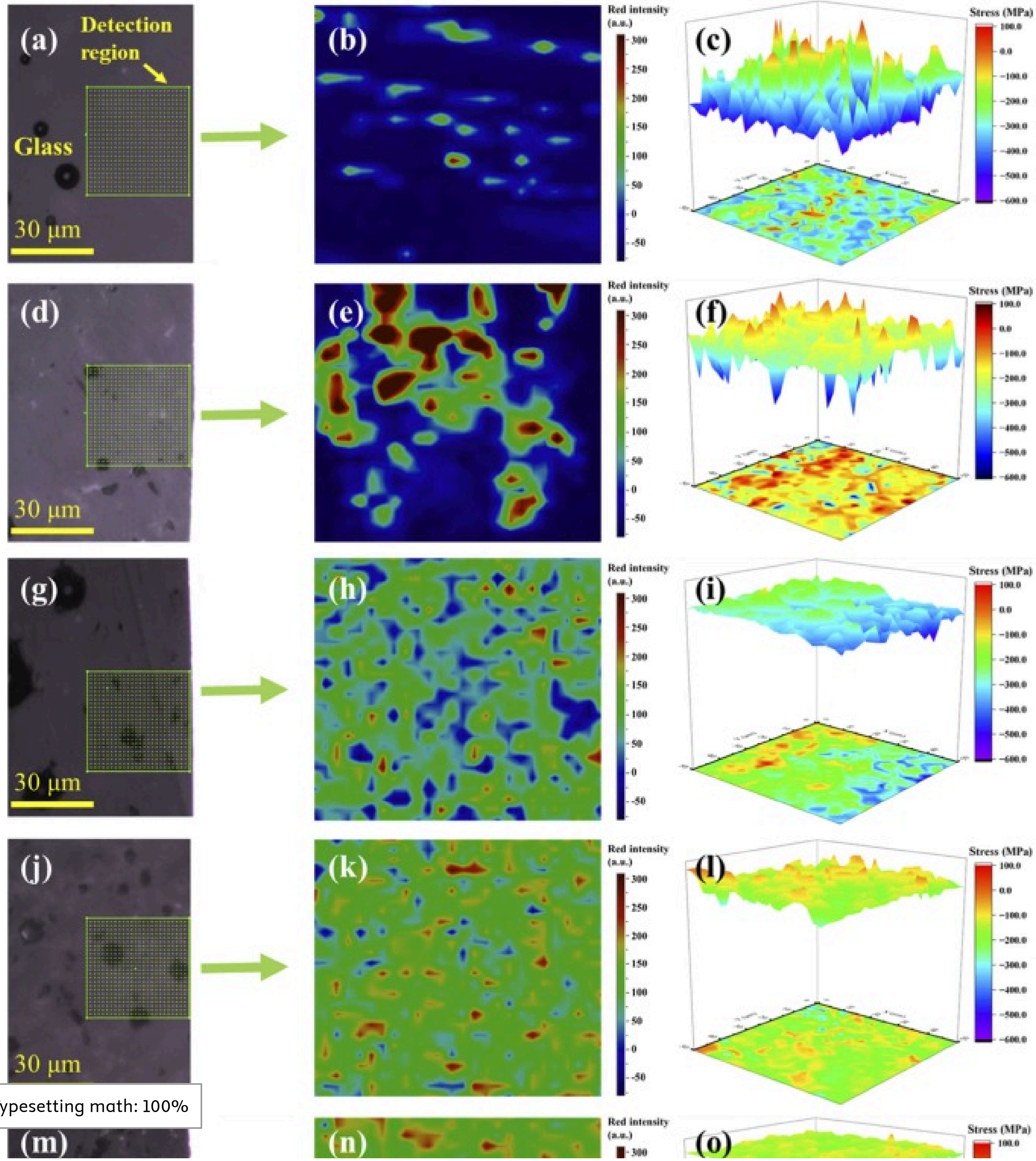
To further elucidate the underlying causes of the observed fluctuations in the measurements, a more detailed surface scan of the samples was conducted with a scanning step of 2 μm. The results are illustrated in Fig. 3**.** Fig. 3a, d, g, j, and m display the actual images of the finely measured regions within the glass, while the corresponding distributions of fluorescence intensity are depicted in Fig. 3b, e, h, k, and n. From the distribution of residual stress and the trends in numerical oscillations, it becomes clear that as the alumina particle content increases, the macroscopic distribution of residual stress fields gradually tends towards uniformity. Notably, the A_3*-unseal* and A_5*-unseal* samples exhibit substantial variations in stress values, with numerous prominent columnar features visible in the three-dimensional contour stress distribution plots (Fig. 3c and f). In contrast, the A_{10}*-unseal*, A_{15}*-unseal* and A_{20}*-unseal* samples (Fig. 3i, l, and o) demonstrate a remarkably uniform stress field distribution with minimal fluctuations.



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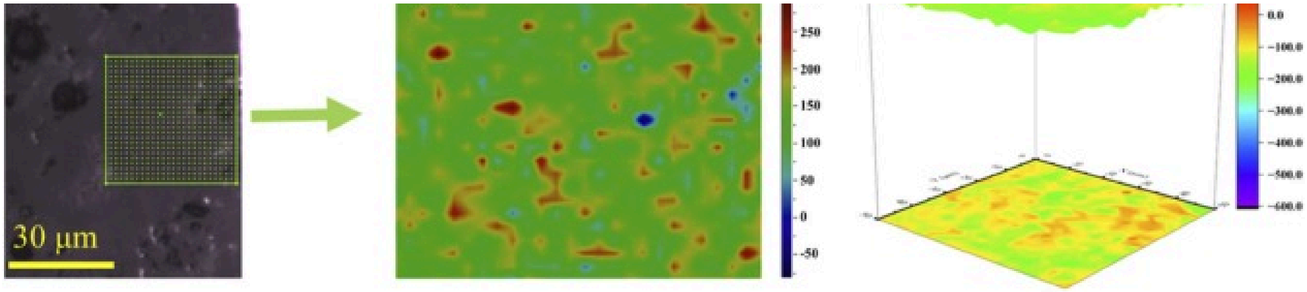

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Fig. 3. Optical microscopy micrograph of the glass region with a grid diagram over a 50 × 50 μm^2 area marked for mapping scanning [(a), (d), (g), (j), (m)]; relative fluorescence intensity of alumina in the scanning area [(b), (e), (h), (k), (n)]; 3D contour plot of the stress field in the glass region [(c), (f), (i), (l), (o)]. (a), (b), (c) represent A_3-*unseal*; (d), (e), (f) represent A_5-*unseal*; (g), (h), (i) represent A_{10}-*unseal*; (j), (k), (l) represent A_{15}-*unseal*; and (m), (n), (o) represent A_{20}-*unseal*, respectively.

The noise observed in the measured values is likely due to the similarity in size between the apertures used in PLS and the pores within the glass region. These pores, which are a result of powder compression during the preform manufacturing process [46], may cause defocusing when collecting PLS data under the microscope. Additionally, the presence of these pores may impact the local stress field in their vicinity [47]. Furthermore, as the alumina content increases, improved contact between neighboring particles result in the overlapping of their respective stress fields. Consequently, stress values measured in samples with higher alumina particle content are more likely to reflect averaged stress fields among adjacent particles, leading to more consistent measurments.

Fig. 4 displays a comparison between experimental detections and theoretical predictions of intrinsic residual stress in samples with varying alumina content. The average detected stress values for A_3-*unseal*, A_5-*unseal*, A_{10}-*unseal*, A_{15}-*unseal*, and A_{20}-*unseal* are recorded as −166 MPa, −149 MPa, −107 MPa, −53 MPa, and −43 MPa, respectively. Correspondingly, the theoretical predictions are −200 MPa, −168 MPa, −133 MPa, −114 MPa, and −100 MPa, respectively. The detected intrinsic residual stress values of each sample with differing alumina contents align closely with the calculated results from the theoretical prediction.

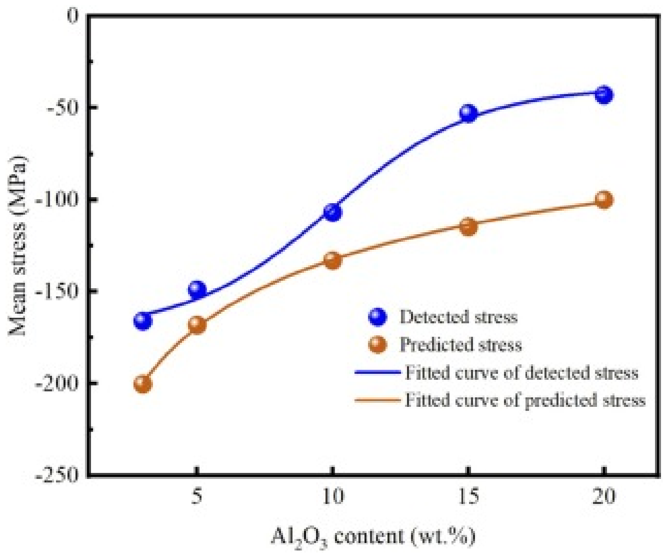

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Fig. 4. Comparison between detected and predicted values for intrinsic residual stress.

It is worth noting that the discrepancies between the experimental results and theoretical calculations may stem from the assumptions of uniform material properties and idealized boundary conditions in the theoretical model, which may not fully capture the complexities of actual material behavior in GMCs. The inherent material inhomogeneity and microstructural defects further contribute to localized stress variations, making accurate predictions challenging. Furthermore, a notable sharp reduction in stress is observed between the A_{10}-*unseal* and A_{15}-*unseal* samples in the empirical results, which may be attributed to overlapping stress fields caused by the higher alumina content. Additionally, the irregular arrangement of Al_2O_3 particles within

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an significantly influence stress distribution along various spatial directions in the composite material. This irregularity may affect the stress approximation based on the average field, providing an explanation for the discrepancies observed between experimental measurements and model predictions [36], [48].

3.2.2. Extra compressive stress

Direct measurement of extra compressive stress is technically unfeasible [49]. Therefore, FEA simulation was employed as an alternative method to determine σ_e. The simplified model used in the FEA simulations, with its geometric structure, is depicted in Fig. 5a, which illustrates the refinement of the mesh. Fig. 5b illustrates the FEA simulated stress results, using the A_3-*seal* as an example. The simulation clearly shows that the stress exerted by the metal housing on the glass region is compressive in nature. For the simulation results of different samples, stress values were taken radially outward from the center of the glass region, as indicated by the arrow in Fig. 5b. The results, as displayed in Fig. 5c, reveal that with an increase in alumina content from A_3-*seal* to A_{20}-*seal*, the extra compressive stress also progressively rises. This increase is primarily attributed to the addition of alumina, which decreases the CTE and increases the elastic modulus of the GMCs, consistent with the predictions of Eq. 8. It is noteworthy that the magnitudes of extra compressive stress obtained from the simulations across samples with varying alumina contents are similar. This similarity can be ascribed to the minimal differences in CTEs and elastic moduli among the samples, as shown in Table 1, resulting in only minor variations in the values of extra compressive stress.

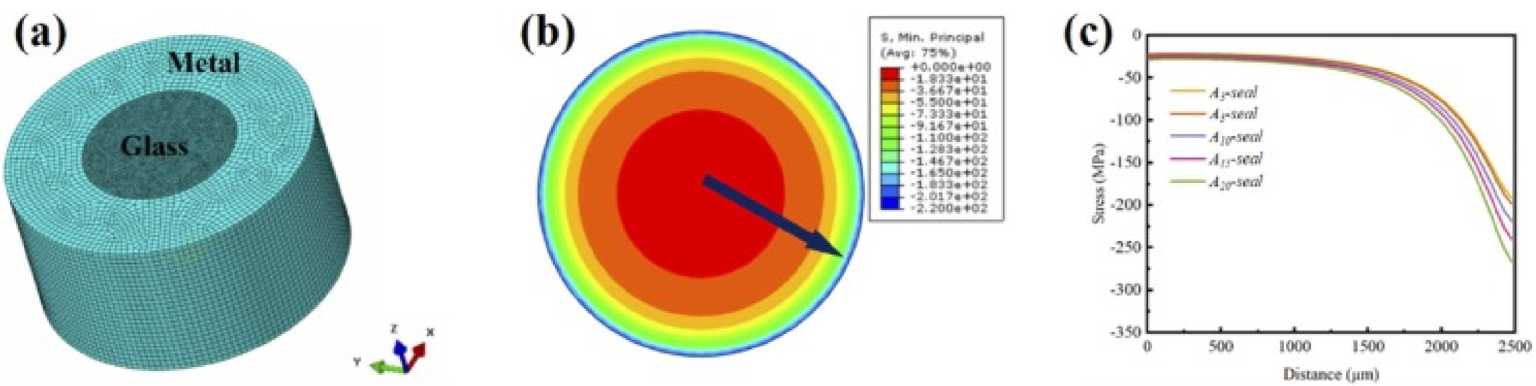

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Fig. 5. (a) Geometry of the FEA simulation model for the GTM seal, (b) Stress distribution within the glass region, (c) Simulation

Typesetting math: 100% mpressive stress for different samples.

To validate the accuracy of the FEA simulations, a comparison was conducted between the stresses at the interface of different samples as obtained from simulations and those predicted theoretically. Notably, for the extra compressive stress at the interface (σ'_e), a theoretical prediction formula was derived based on the assumption of equal strain between the metal housing and the glass, which is presented as follows [50]:

$$\alpha_{gc}\Delta T + \frac{2\sigma'_e}{E_{gc}} = \alpha_m \Delta T + \frac{2\sigma'_e}{E_m} \tag{9}$$

Consequently, the value of σ'_e can be calculated using the following equation:

$$\sigma'_e = \frac{E_{gc}E_m(\alpha_{gc}-\alpha_m)\Delta T}{2(E_{gc}+E_m)} \tag{10}$$

Fig. 6 illustrates a comparison between the FEA simulation results and theoretical predictions of extra compressive stress at the interface for samples with different alumina concentrations. The FEA simulation results for the A_3-*seal*, A_5-*seal*, A_{10}-*seal*, A_{15}-*seal*, and A_{20}-*seal* are −191 MPa, −198 MPa, −218 MPa, −240 MPa, and −266 MPa, respectively. The corresponding theoretical predictions are −167 MPa, −171 MPa, −182 MPa, −196 MPa, and −213 MPa, respectively. Notably, the average value of the extra compressive stress obtained by solving (8), (9), (10) is presented in , Appendix A Supplementary material of the supplementary materials, with the specific values shown in , Appendix A Supplementary material. Both the stress at the interface and the average extra compressive stress exhibits a consistent trend, providing better insights into the reliability of the results.

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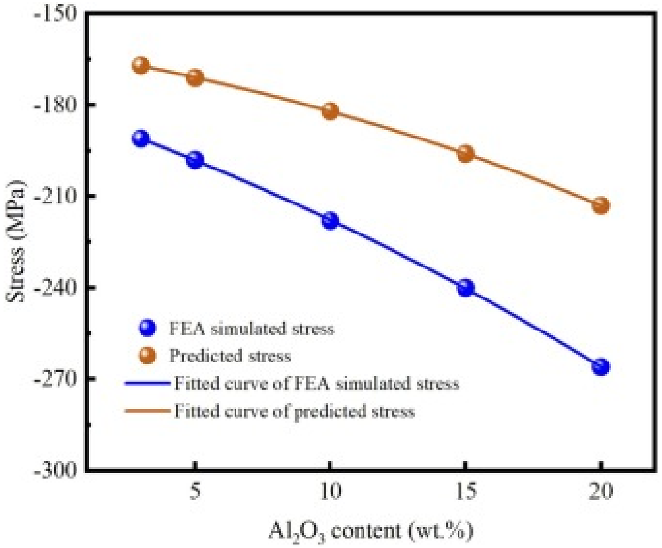

Fig. 6. Comparison between FEA simulated and predicted values for the extra compressive stress.

Despite the slight discrepancies observed, which may arise from the fact that FEA simulations incorporate a more detailed representation of the stress distribution, taking into account localized effects, such effects can lead to variations in stress distribution that are not captured by the simplified theoretical model. In addition, potential structural relaxation processes during the sealing procedure may not be precisely accounted for, contributing to the observed differences. However, the close alignment of the compressive stress values obtained from FEA simulations with the theoretical predictions for samples with different alumina contents substantiates the accuracy of the finite element simulations. Thus, these simulation results can be reliably employed to represent the extra compressive stress exerted by the metal housing.

3.2.3. Total residual stress

Fig. 7a illustrates the total residual stress outcomes for GTM seals with diverse contents of Al_2O_3, ascertained through the PLS method. Commencing from the center, the Raman laser methodically extended its scan to the periphery of the glass region.

Typesetting math: 100% m step, it encompassed a span of around 2500 μm, thereby charting the radial residual stress profile of the glass region within the seals. The results can be categorized into four distinct zones. In Zone I (0–250 μm), a significant reduction in compressive stress is observed as the radial distance x increases, which is primarily attributed to the intrinsic contraction of the glass during cooling. Although minor variations are present, Zone II (250–1250 μm) shows a steady decline in compressive stress, predominantly influenced by the thermal gradient extending from the core glass to the interface. Here, the glass in Zone I exhibits higher viscosity during cooling compared to Zone II. As the central glass (Zone I) remains molten while other zones solidify sooner, the slower contraction in Zone I induces tensile stress on the adjacent rigid glass, thereby diminishing the compressive stress in Zone II. Zone III (1250–2150 μm) demonstrates a slight increase in stress with x, where variations in compressive stress may be associated with bubble formation or irregularities in the glass structure. Notably, in Zone IV, a pronounced surge in compressive stress primarily stems from the rapid contraction of the metal housing during cooling, with the stress effect limitted to within 350 μm of the glass-metal interface [27]. Interestingly, in the A_{15}-*seal* and A_{20}-*seal* samples, the abrupt increase in compressive stress at the interface is significantly less, possibly due to variations in temperature gradients.

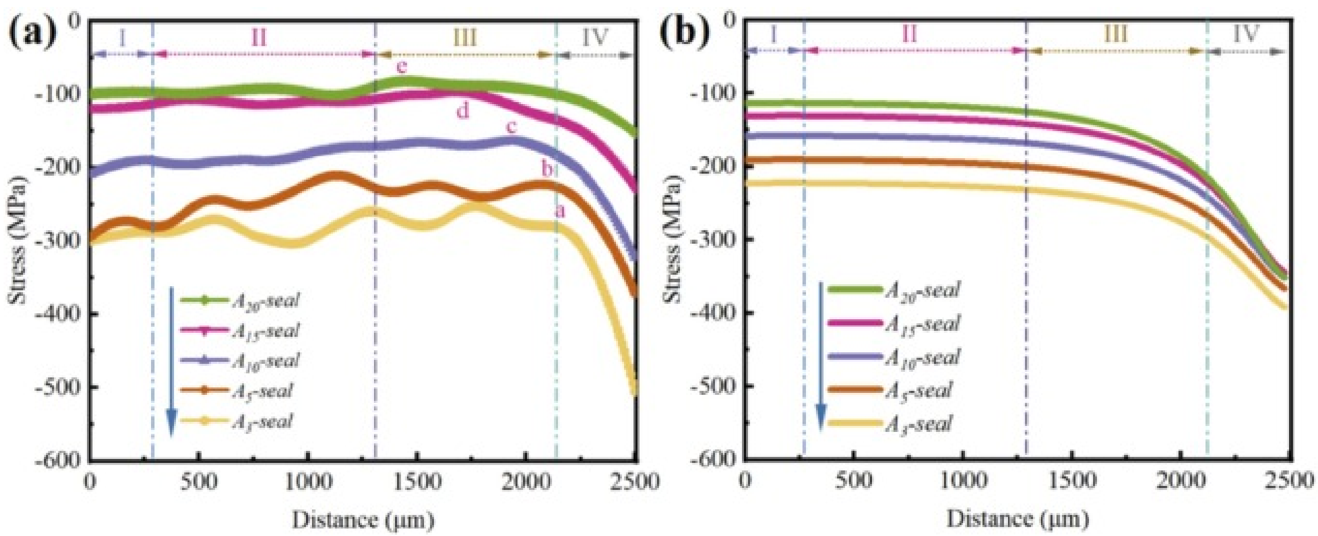

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Fig. 7. Total residual stress obtained from PLS measurements (a) and predicted results (b).

A noteworthy feature in the stress distribution diagrams is the consistent presence of a “turning point,” marked as a, b, c, d, and e in Fig. 7a. The “turning point” indicates a sharp increase in stress values. It is evident that as the content of Al_2O_3 particles

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increases, the “turning point” gradually shifts leftwards, i.e., closer to the center. Specifically, the “turning point” for the A_3*-seal*, A_5*-seal*, A_{10}*-seal*, A_{15}*-seal*, and A_{20}*-seal* are respectively situated at 2130 μm, 1880 μm, 1940 μm, 1760 μm, and 1410 μm from the central area. This trend is predominantly driven by the enhanced thermal conductivity of the GMCs due to the Al_2O_3 particle integration, fostering a more homogeneous temperature distribution during cooling [51], [52]. Consequently, this results in a decreased internal temperature gradient within the glass, leading to a more uniform contraction and the inward migration of the compressive stress “turning point.”

Fig. 7b presents the predicted results of the total residual stresses for different samples, derived from Eq. 1, with corresponding Zones I, II, III, and IV also highlighted. It is observed that the theoretical predictions for Zones I and III align well with the empirical data, whereas some discrepancies are noted in Zone II. These inconsistencies may stem from non-uniformities in the temperature field within the samples, as opposed to the uniform conditions assumed in the finite element model. Notably, in Zone IV, both the empirical results and theoretical predictions exhibit a sharp increase in stress, primarily attributed to the intense inward compression exerted by the metal housing.

To evaluate the aligment between the experimentally detected results and the theoretically predicted calculations, arithmetic averages of the measured and predicted total residual stresses were computed. The results are displayed in Fig. 8. Notably, to minimize the effects of temperature gradients, stress values from area IV were not considered. The measured stress values for A_3*-seal*, A_5*-seal*, A_{10}*-seal*, A_{15}*-seal*, and A_{20}*-seal* are −279 MPa, −225 MPa, −169 MPa, −112 MPa, and −97 MPa, respectively. The corresponding theoretical predictions are −261 MPa, −239 MPa, −184 MPa, −166 MPa, and −151 MPa, respectively.

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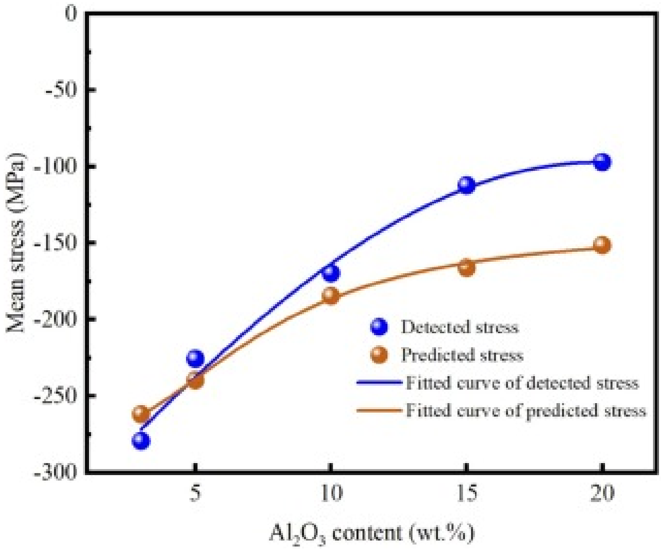

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Fig. 8. Comparison between detected and predicted values for the total residual stress.

The measured value of *A_3-seal* is slightly higher than its predicted value, which might be attributed to experimental errors stemming from significant fluctuations in the measurements. Apart from this, the measured values for the other samples were all lower than their predicted values. The observed discrepancies between the experimental and predicted total residual stress values can be attributed to several key factors inherent in the actual sealing process. First, the inherent variability in material properties, such as the distribution and orientation of alumina particles within the glass matrix, can lead to variations in stress that are difficult to fully capture through theoretical models. What's more, the presence of temperature gradients during the sealing process and potential structural relaxation effects are complex phenomena that cannot be comprehensively considered in theoretical calculations. These factors, combined with potential noise interference during experimental measurements, may contribute to the differences observed between the experimental and predicted stress values. However, it is noteworthy that despite these discrepancies, the overall trend observed in the experimental data aligns closely with the theoretical predictions. Specifically, the total residual stress values measured experimentally using the PLS method consistently decrease with increasing

which is a key finding of this study. This trend is crucial, as it confirms the expected behavior and validates the impact of alumina content on reducing residual stress in GTM seals.

3.3. Alumina content-stress relationship

The theoretical predictions and experimental measurements clearly indicate that the total residual stress in GTM seals exhibit a trend of decreasing with increasing alumina content. To further elucidate the relationship between the alumina content and the stress, an analysis of strain energy within the GMCs during the cooling process was conducted. This analysis aims to provide a deeper understanding of the spatial distribution of stresses influenced by varying concentrations of alumina particles. To simplify the calculations, a spherical coordinate system was employed. Fig. 9 illustrates the computational schematic, where it is assumed that the matrix is infinitely larger than the inclusions. In this diagram, R_1 represents the initial size of the inclusion particles, r_1 denotes the size of the inclusion particles after being compressed by the matrix during the cooling process, and R_2 represents the radius of the matrix.

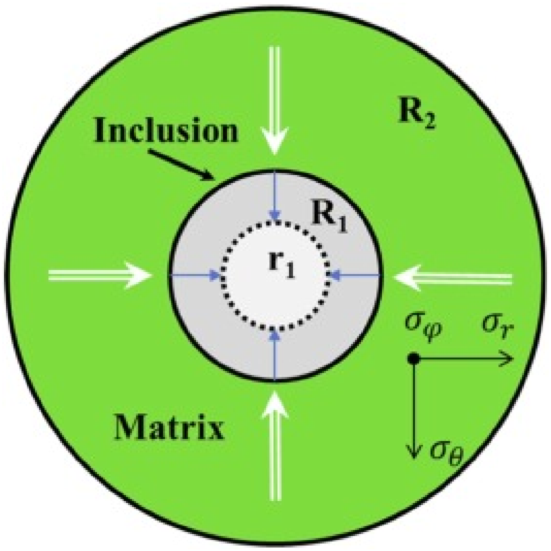

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Fig. 9. Schematic diagram illustrating the calculation of stress, strain, and strain energy around an inclusion.

The r_1 can be calculated through the following equation [53]:

$$\left\{3\ln\frac{R_1}{r_1}+\left[1-\left(\frac{R_1}{R_2}\right)^3\right]\right\}=\frac{3}{2}\left[(\alpha_i-\alpha_m)\frac{\Delta T}{Y}+\left(\frac{1-v_m}{E_m}\right)\left(\frac{R_1}{r_1}\right)^3\right] \quad (11)$$

Where, E denotes the Young's moduli, and ν is the Poisson's ratio. The subscripts m and i correspond to the matrix and inclusion, respectively. Y is the compressive strength of the matrix, which is a fixed value specific to the material.

In spherical coordinates, the relationship between displacement u and strain ε is as follows:

$$\varepsilon_r = \frac{du}{dr} \tag{12}$$

$$\varepsilon_\theta = \varepsilon_\varphi = \frac{u}{r} \tag{13}$$

It can be obtained by combining the elastic constitutive equation and solving the second order differential equation about u and r [53]:

$$\varepsilon_r = \frac{1}{E}(\sigma_r - 2\upsilon_m\sigma_\theta) \tag{14}$$

$$\varepsilon_\theta = \frac{1}{E}[(1-\upsilon_m)\sigma_\theta - \upsilon_m\sigma_r] \tag{15}$$

Thus, the strain energy in the inclusion, $E_{s,i}$, can be obtained from the following equation:

$$E_{s,i} = \frac{1}{2}(\sigma_r\varepsilon_r + 2\sigma_\theta\varepsilon_\theta) = \frac{1}{2E}[\sigma_r^2 + 2(1-\upsilon_m)\sigma_\theta^2 - 4\upsilon_m\sigma_r\sigma_\theta] \tag{16}$$

Furthermore, by substituting the boundary condition $r_1 = R_1$ and solving for the expression of u, the following can be obtained:

$$E_{s,i} = \frac{4Y^2}{3E}\left[(1-2\nu)h^3 + \frac{1+\nu}{2}\right]\left(\frac{R_1}{r_1}\right)^3 \tag{17}$$

The strain energy $E_{s,g}$, generated within the glass matrix can be expressed as follows:

$$E_{s,g} = \frac{1}{2}\int \sigma_g\varepsilon_g\, dV \tag{18}$$

Where, ε_g represents the strain, and σ_g represents the stress in the glass matrix. σ_g can be calculated using the equilibrium condition [36]:

$$\sigma_g = \frac{-f\sigma_i}{1-f} \tag{19}$$

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Referring to Eq. 18, for GMCs containing Al_2O_3 particles, assuming an infinitely large glass matrix, the strain energy generated during the cooling process remains constant. Fig. 10a and b depict schematic representations of the strain energy field distribution within the glass matrix during cooling for composites with low and high inclusion contents, respectively. When the alumina content in the composite is relatively low, a significant portion of the strain energy within the matrix is concentrated on specific particles, leading to a reduction in r_1, as evident from Eq. 17. This indicates more substantial deformation for each particle, and according to Hooke's law, σ=ε·E, this leads to higher σ_i values. In contrast, for GMCs with higher alumina content, each particle acquires less strain energy, resulting in smaller deformations of the particles, namely, $r_1' > r_1$, which leads to lower σ_i values. As noted in Section 3.2.1, lower concentrations of Al_2O_3 particles correspond to higher intrinsic residual stress, while higher concentrations result in lower intrinsic residual stress.

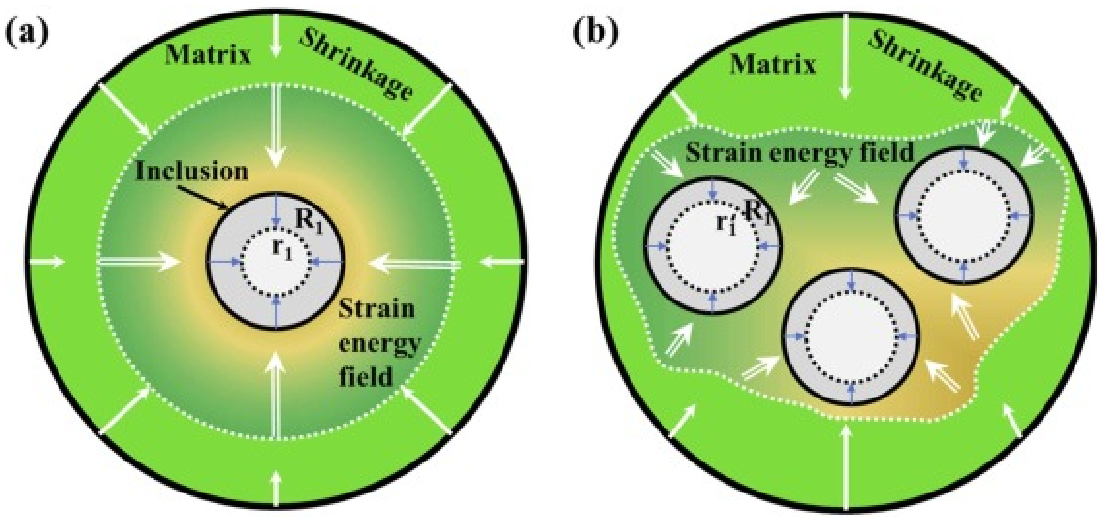

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Fig. 10. Schematic depiction of strain energy field distribution in composite materials with inclusions during the cooling process: (a) low inclusion content, (b) high inclusion content.

In the case of GTM seals components, the introduction of extra compressive stress σ_e from the contraction of the metal housing is also considered. However, it is important to observe that, as demonstrated in the findings of Section 3.2.2, the magnitude of this

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[…]e stress remains nearly consistent across samples with different alumina contents. Hence, while the introduction of extra compressive stress may further amplify the strain energy within the glass matrix, it does not significantly alter the distribution of the strain energy field within the composite. In composites with low alumina content, each particle receives a larger share of strain energy, resulting in increased deformation, and consequently, greater stress. Conversely, in glass composites with high alumina content, each particle absorbs less strain energy, leading to reduced deformation and lower stress. The total residual stress outcomes, as detailed in Section 3.2.3, are in close alignment with this analysis. To further clarify and validate the mechanism discussed above, FEA simulations were performed on GMCs-based GTM seals with varying alumina content. The resulting particle strain data are presented in , Appendix A Supplementary material in the supplementary materials. These results are consistent with the previous analysis, further confirming the validity and accuracy of this explanation.

4. Conclusion

To explore the residual stress state and its underlying mechanisms in GTM seals fabricated with GMCs containing varying amounts of alumina, Cr^{3+}-doped Al_2O_3 particles were employed, and PLS was used for precise stress measurement. By integrating theoretical calculations, experimental characterization, and FEA simulations, a comprehensive understanding of the residual stress in GTM seals was achieved. The key conclusions are as follows:

(1) Residual stress in GTM seals fabricated with GMCs can be classified into two distinct types: intrinsic residual stress, arising from the mismatch in CTE between the glass matrix and alumina, and extra compressive stress, resulting from the cooling-induced contraction of the metal housing. The total residual stress in GTM seals is the sum of these two components;

(2) Experimental results showed that as the alumina content increased, both intrinsic and total residual stress decreased substantially, consistent with theoretical predictions. FEA simulations further confirmed this trend;

(3) Strain energy theory analysis revealed that increasing the alumina content within the composite decreases the energy transferred to each particle, leading to reduced deformation and corresponding stress;

(4) Alumina particles significantly enhance stress field homogeneity in the composites. Higher alumina content results in a more uniform stress distribution across the GTM seals and modifies the thermal gradient during cooling, further improving stress uniformity in the glass region.

This analysis of residual stress in GMC-based GTM seals offers valuable insights for optimizing the fabrication of composite GTM seals. The application of PLS technology serves as a powerful tool for studying residual stress, while the introduction of strain energy theory advances the understanding of stress variation mechanisms. These findings not only improve GTM seals design but also have broad applicability to other composite material systems.

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CRediT authorship contribution statement

Keqian Gong: Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Zheng Liu:** Visualization, Validation, Investigation. **Yangyang Cai:** Visualization. **Zifeng Song:** Methodology. **Chao Zhou:** Visualization. **Jing Liu:** Visualization. **Yuna Zhao:** Validation. **Yong Zhang:** Writing – review & editing, Validation, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

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Supplementary material

Recommended articles

Data availability

Data used in this manuscript will be made available by the corresponding author upon reasonable request.

References

[1] T.U. Shinde, V.H. Dalvi, R.G. Patil, C.S. Mathpati, S.V. Panse, J.B. Joshi
Thermal performance analysis of novel receiver for parabolic trough solar collector
Energy, 254 (2022), 10.1016/j.energy.2022.124343
Google Scholar

[2] Y.Y. Cai, K.J. Hu, S.H. Li, Q.Y. Zhu, Z. Liu, K.Q. Gong, Y. Zhang
Structure optimization of electrical penetration assemblies: experimental facts and stress field analysis by the finite element method
Materialia, 22 (2022), 10.1016/j.mtla.2022.101381
Google Scholar

[3] X. Li, S.M. Groß-Barsnick, T. Koppitz, S. Baumann, W.A. Meulenberg, G. Natour
Sealing behaviour of glass-based composites for oxygen transport membranes
J. Eur. Ceram. Soc., 42 (2022), pp. 2879-2891, 10.1016/j.jeurceramsoc.2022.01.052
View PDF View article View in Scopus Google Scholar

[4] X. Wang, X. Gao, Z. Zhang, L. Cheng, H. Ma, W. Yang
Advances in modifications and high-temperature applications of silicon carbide ceramic matrix composites in aerospace: a focused review
J. Eur. Ceram. Soc., 41 (2021), pp. 4671-4688, 10.1016/j.jeurceramsoc.2021.03.051
View PDF View article View in Scopus Google Scholar

Typesetting math: 100%

[5] I.W. Donald, P.M. Mallinson, B.L. Metcalfe, L.A. Gerrard, J.A. Fernie

Recent developments in the preparation, characterization and applications of glass- and glass–ceramic-to-metal seals and coatings
J. Mater. Sci., 46 (2011), pp. 1975-2000, 10.1007/s10853-010-5095-y
View in Scopus Google Scholar

[6] J. Yue, X. Chen, Y. Garbatov
Ultimate strength assessment of penetrated nuclear reactor containment vessel subjected to external pressure and thermal loads
Ocean Eng., 273 (2023), Article 113975, 10.1016/j.oceaneng.2023.113975
View PDF View article View in Scopus Google Scholar

[7] C. Chanmuang, M. Naksata, T. Chairuangsri, H. Jain, C.E. Lyman
Microscopy and strength of borosilicate glass-to-Kovar alloy joints
Mater. Sci. Eng. A, 474 (2008), pp. 218-224, 10.1016/j.msea.2007.04.016
View PDF View article View in Scopus Google Scholar

[8] I.W. Donald
Preparation, properties and chemistry of glass- and glass-ceramic-to-metal seals and coatings
J. Mater. Sci., 28 (1993), pp. 2841-2886, 10.1007/bf00354689
View in Scopus Google Scholar

[9] C. Pascal, M. Braccini, V. Parry, E. Fedorova, M. Mantel, D. Oquab, D. Monceau
Relation between microstructure induced by oxidation and room-temperature mechanical properties of the thermally grown oxide scales on austenitic stainless steels
Mater. Charact., 127 (2017), pp. 161-170, 10.1016/j.matchar.2017.03.003
View PDF View article View in Scopus Google Scholar

[10] R. Li, X. Liang, X. Wang, W. Zeng, J. Yang, D. Yan, J. Pu, B. Chi, J. Li
Improvement of sealing performance for Al_2O_3 fiber-reinforced compressive seals for intermediate Typesetting math: 100% ure solid oxide fuel cell
Ceram. Int., 45 (2019), pp. 21953-21959, 10.1016/j.ceramint.2019.07.209

View PDF View article View in Scopus Google Scholar

[11] S. Li, Y. Cai, Q. Zhu, Z. Liu, Y. Zhang
Interface degradation of glass-to-metal seals during thermo-oxidative aging
Corros. Sci., 199 (2022), Article 110189, 10.1016/j.corsci.2022.110189
View PDF View article View in Scopus Google Scholar

[12] Z. Fan, X. Diao, M. Liu, Y. Zhang, Z. Huang, H. Yan
On-line monitoring of sealing glass in electrical penetration assembly based on femto-laser inscribed fiber Bragg grating sensors
Opt. Express, 27 (2019), pp. 608-620, 10.1364/oe.27.000608
View in Scopus Google Scholar

[13] B. Deng, J. Luo, J.T. Harris, C.M. Smith
Critical stress map for ZrO_2 tetragonal to monoclinic phase transformation in ZrO_2-toughened glass-ceramics
Materialia, 9 (2020), Article 100548, 10.1016/j.mtla.2019.100548
View PDF View article View in Scopus Google Scholar

[14] L. Liu, F. Ye, Y. Zhou, Z. Zhang
Microstructure compatibility and its effect on the mechanical properties of the α-SiC/β-Si_3N_4 co-reinforced barium aluminosilicate glass ceramic matrix composites
Scr. Mater., 63 (2010), pp. 166-169, 10.1016/j.scriptamat.2010.03.040
View PDF View article View in Scopus Google Scholar

[15] M.J. Pascual
Glasses and glass-ceramics as sealing materials
Encyclopedia of Materials: Technical Ceramics and Glasses, 2 (2021), pp. 746-761, 10.1016/B978-0-12-818542-1.00047-3
View PDF View article View in Scopus Google Scholar

[16] Q.G. Jiang, C. Cao, T.C. Lin, S. Wu, X. Li
Typesetting math: 100% d tough glass with self-dispersed nanoparticles via solidification

Adv. Mater., 31 (2019), p. 1901803, 10.1002/adma.201901803 ↗

View in Scopus ↗ Google Scholar ↗

[17] J.W. Zwanziger, U. Werner-Zwanziger, E.D. Zanotto, E. Rotari, L.N. Glebova, L.B. Glebov, J.F. Schneider
Residual internal stress in partially crystallized photothermorefractive glass: evaluation by nuclear magnetic resonance spectroscopy and first principles calculations
J. Appl. Phys., 99 (2006), Article 083511, 10.1063/1.2191731 ↗

View in Scopus ↗ Google Scholar ↗

[18] C.-H. Hsueh, P.F. Becher
Residual thermal stresses in ceramic composites. Part I: with ellipsoidal inclusions
Mater. Sci. Eng. A, 212 (1996), pp. 22-28, 10.1016/0921-5093(96)10176-3 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[19] J. Selsing
Internal stresses in ceramics
J. Am. Ceram. Soc., 44 (1961), pp. 419-4119, 10.1111/j.1151-2916.1961.tb15475.x ↗

View in Scopus ↗ Google Scholar ↗

[20] F.W. Martin
Stresses in Glass-Metal seals: I, the cylindrical seal
J. Am. Ceram. Soc., 33 (1950), pp. 224-229, 10.1111/j.1151-2916.1950.tb14911.x ↗

View in Scopus ↗ Google Scholar ↗

[21] Z. Liu, Y. Cai, K. Gong, W. Zhou, F. Chen, Y. Zhang
Calibration for determination of compressive stress in glass-to-metal seals via photoluminescence spectroscopy
J. Am. Ceram. Soc., 107 (2023), pp. 1219-1231, 10.1111/jace.19528 ↗

Google Scholar ↗

Typesetting math: 100% , J. Wang, J. Qi, J. Zhang, Z. Yin

Analysis of residual stress for mismatch metal–glass seals in solar evacuated tubes
Sol. Energy Mater. Sol. Cells, 128 (2014), pp. 421-426, 10.1016/j.solmat.2014.05.042 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[23] F.C. Serbena, E.D. Zanotto
Internal residual stresses in glass-ceramics: a review
J. Non Cryst. Solids, 358 (2012), pp. 975-984, 10.1016/j.jnoncrysol.2012.01.040 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[24] P.J. Withers, H.K.D.H. Bhadeshia
Residual stress. Part I – measurement
Tech., J. Mater. Sci. Technol., 17 (2013), pp. 355-365, 10.1179/026708301101509980 ↗
Google Scholar ↗

[25] K. Hu, H. Yan, Z. Fan, X. Diao, M. Liu, S. Li, Y. Zhang
In situ characterization of residual stress in glass-to-metal seal
Ceram. Int., 45 (2019), pp. 20983-20987, 10.1016/j.ceramint.2019.06.259 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[26] K. Zeng, D.J. Rowcliffe
Vickers indentations in glass—I. Residual stress fields and iso-stress contour maps
Acta Met., 43 (1995), pp. 1935-1943, 10.1016/0956-7151(94)00392-u ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[27] S. Li, Q. Zhu, K. Hu, Y. Cai, Z. Liu, F. Chen, Y. Zhang
Determination of compressive stress in glass-to-metal seals using photoluminescence spectroscopy technique
Ceram. Int., 48 (2022), pp. 13379-13385, 10.1016/j.ceramint.2022.01.219 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

Typesetting math: 100% Clarke

Stress measurement in single-crystal and polycrystalline ceramics using their optical fluorescence
J. Am. Ceram. Soc., 76 (1993), pp. 1433-1440, 10.1111/j.1151-2916.1993.tb03922.x ↗
View in Scopus ↗ Google Scholar ↗

[29] Y.B. Gerbig, S.J. Stranick, R.F. Cook
Measurement of residual stress field anisotropy at indentations in silicon
Scr. Mater., 63 (2010), pp. 512-515, 10.1016/j.scriptamat.2010.05.017 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[30] P. Zorabedian, F. Adar
Measurement of local stress in laser-recrystallized lateral epitaxial silicon films over silicon dioxide using Raman scattering
Appl. Phys. Lett., 43 (1983), pp. 177-179, 10.1063/1.94271 ↗
View in Scopus ↗ Google Scholar ↗

[31] C. Galiotis, A. Paipetis, C. Marston
Unification of fibre/matrix interfacial measurements with Raman microscopy
J. Raman Spectrosc., 30 (1999), pp. 899-912, 10.1002/(sici)1097-4555(199910)30:10<899::Aid-jrs465>3.0.Co;2-v ↗
View in Scopus ↗ Google Scholar ↗

[32] K.W. Schlichting, K. Vaidyanathan, Y.H. Sohn, E.H. Jordan, M. Gell, N.P. Padture
Application of Cr^{3+} photoluminescence piezo-spectroscopy to plasma-sprayed thermal barrier coatings for residual stress measurement
Mater. Sci. Eng. A, 291 (2000), pp. 68-77, 10.1016/s0921-5093(00)00973-4 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[33] R.F. Cook, C.A. Michaels
Review: coefficients for stress, temperature, and composition effects in fluorescence measurements of alumina
st. Stand Technol., 122 (2017), pp. 1-26, 10.6028/jres.122.043 ↗
Google Scholar ↗

[34] E. Huntley, K. Strong, B. Elisberg, S. Meserole, T. Diebold
Photoluminescence spectroscopy to determine residual stresses in glass-to-metal seals
Off. Sci. Tech. Inf. (OSTI) (2019), 10.2172/1592889
Google Scholar

[35] R.J. Young, X. Yang
Interfacial failure in ceramic fibre/glass composites
Compos. Part A, 27 (1996), pp. 737-741, 10.1016/1359-835x(96)00022-x
View PDF View article View in Scopus Google Scholar

[36] R.I. Todd, A.R. Boccaccini, R. Sinclair, R.B. Yallee, R.J. Young
Thermal residual stresses and their toughening effect in Al_2O_3 platelet reinforced glass
Acta Mater., 47 (1999), pp. 3233-3240, 10.1016/s1359-6454(99)00177-9
View PDF View article View in Scopus Google Scholar

[37] C.A. Michaels, R.F. Cook
Determination of residual stress distributions in polycrystalline alumina using fluorescence microscopy
Mater. Des., 107 (2016), pp. 478-490, 10.1016/j.matdes.2016.06.063
View PDF View article View in Scopus Google Scholar

[38] K. Gong, Z. Song, Y. Cai, Z. Liu, Z. Shi, C. Zhou, H. Yan, Y. Zhang
Assessment of residual stress evolution in glass-to-metal seals amid heating process: Insights from in situ observations and finite-element analysis
J. Appl. Phys., 136 (2024), Article 075101, 10.1063/5.0214312
View in Scopus Google Scholar

[39] S. Li, K. Hu, W. Hui, Y. Cai, Q. Zhu, Y. Zhang
Morphological evolution of oxide layer and its effect on glass-to-metal seal
J. Non-Cryst. Solids, 549 (2020), Article 120355, 10.1016/j.jnoncrysol.2020.120355
F View article View in Scopus Google Scholar

Typesetting math: 100%

[40] Y. Cai, Z. Liu, K. Gong, Y. Zhang
The effect of reinforcement particle size on the mechanical and fracture properties of glass matrix composites
Heliyon, 9 (2023), Article e21895, 10.1016/j.heliyon.2023.e21895 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[41] M.M.R.A. Lima, R.C.C. Monteiro, M.P.F. Graça, M.G. Ferreira
da Silva, Structural, electrical and thermal properties of borosilicate glass–alumina composites
J. Alloy. Compd., 538 (2012), pp. 66-72, 10.1016/j.jallcom.2012.05.024 ↗
View PDF View article Google Scholar ↗

[42] E. Bernardo, J. Doyle, S. Hampshire
Sintered feldspar glass–ceramics and glass–ceramic matrix composites
Ceram. Int., 34 (2008), pp. 2037-2042, 10.1016/j.ceramint.2007.07.027 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[43] H. Rawson
The theory of stresses in two-component glass to metal tube seals
J. Instrum., 26 (1949), pp. 25-27, 10.1088/0950-7671/26/1/308 ↗
View in Scopus ↗ Google Scholar ↗

[44] H. Rawson
A theory of stresses in glass butt seals
Br. J. Appl. Phys., 2 (1951), pp. 151-156, 10.1088/0508-3443/2/6/302 ↗
View in Scopus ↗ Google Scholar ↗

[45] A.K. Varshneya
The set point of glass in glass-to-metal sealing
J. Am. Ceram. Soc., 63 (2006), pp. 311-315, 10.1111/j.1151-2916.1980.tb10728.x ↗
Typesetting math: 100% lar ↗

[46] K.Q. Gong, Y.Y. Cai, Z. Liu, Y. Zhang
Thermal cycle stability of glass-to-metal seals with glass preforms produced via powder-metallurgy and casting-machining methods
Mater. Res. Express, 10 (2023), Article 025201, 10.1088/2053-1591/acb641 ↗
View in Scopus ↗ Google Scholar ↗

[47] D.P.H. Hasselman, R.M. Fulrath
Micromechanical stress concentrations in two-phase brittle-matrix ceramic composites
J. Am. Ceram. Soc., 50 (1967), pp. 399-404, 10.1111/j.1151-2916.1967.tb15143.x ↗
View in Scopus ↗ Google Scholar ↗

[48] J.D. Eshelby
The determination of the elastic field of an ellipsoidal inclusion, and related problems
Proc. R. Soc. Lond., Ser. A, 241 (1997), pp. 376-396, 10.1098/rspa.1957.0133 ↗
Google Scholar ↗

[49] D. Lei, Z. Wang, J. Li
The analysis of residual stress in glass-to-metal seals for solar receiver tube
Mater. Des., 31 (2010), pp. 1813-1820, 10.1016/j.matdes.2009.11.012 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[50] D.Q. Lei, Z.F. Wang, Z.J. Wang
Effects of geometry and material properties on the residual stress of glass-to-metal seals in solar receiver tubes
Energy Procedia, 49 (2014), pp. 418-427, 10.1016/j.egypro.2014.03.045 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[51] S. Choi, J. Kim
Thermal conductivity of epoxy composites with a binary-particle system of aluminum oxide and aluminum nitride fillers
Compos. Part B, 51 (2013), pp. 140-147, 10.1016/j.compositesb.2013.03.002 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[52] S. Ren, J. Liu, A. Guo, W. Zang, H. Geng, X. Tao, H. Du
Mechanical properties and thermal conductivity of a temperature resistance hollow glass microspheres/borosilicate glass buoyance material
Mater. Sci. Eng. A, 674 (2016), pp. 604-614, 10.1016/j.msea.2016.08.014 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[53] D.V. Brooksbank, Stress fields around inclusions and their relation to mechanical properties, Defense Technical Information Center, United States, 1972.
Google Scholar ↗

Cited by (5)

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